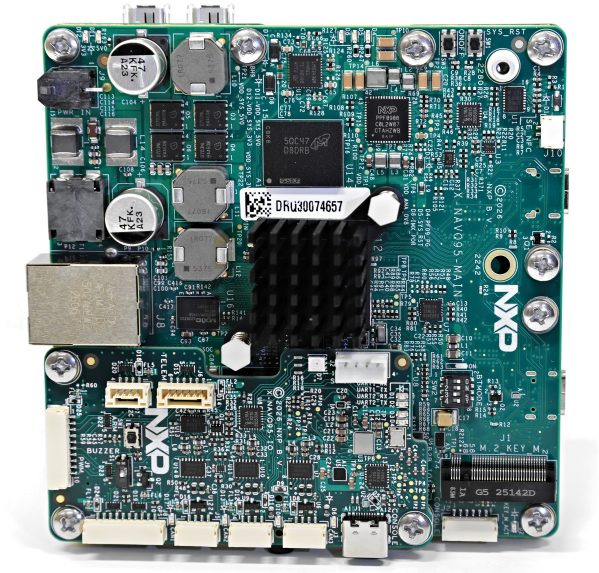
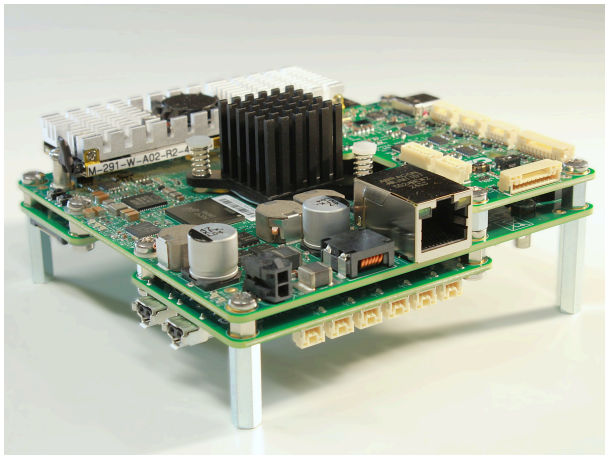
NXP NavQ95 Vehicle Computer

Welcome to the documentation for the NavQ95, a vehicle computer reference design designed around the NXP i.MX95 developed by NXP Mobile Robotics. This board is available from NXP and NXP Distributors worldwide as P/N MR-NAVQ95-KIT and is supported by the NXP Robotics SDK. Accessories for this board will also listed on https://www.NXP.com/MR-NavQ95

The NavQ95 features a single main board with a small form factor and is designed to merge a Vehicle Management Unit and a Vehicle Companion Computer into one unified, heterogeneous MCU/MPU device.

> **Tip**
>
> See MR-NAVQ95/Mr Solutions MR-MR-NAVQ95-RevB description slides (Public).pdf also for a short overview presentation.

MR-NavQ95 Top view	MR-NavQ95 Side View
	

Core Architecture & AI

The heterogeneous vehicle computer is powered by the NXP i.MX 95 processor and is divided into specific execution domains:

- **Compute**: 6x Arm Cortex-A55 cores dedicated to executing compute-intense functions.
- **Real-Time Control**: 1x Arm Cortex-M7 core to execute real-time control tasks.
- **System Management**: 1x Arm Cortex-M33 core to execute the NXP System Manager.

AI Capabilities:

- Equipped with an on-chip NXP Neutron NPU.
- High-bandwidth M.2 PCIe interfaces target up to 2 NXP Kinara Ara240 NPUs.
- Supports cloud-based AI via Wifi6 or 5G connectivity

Software Stack:

The board is designed to run open-source software, leveraging its multi-core architecture:

- Cortex-A55 Cores: Runs the Ubuntu PoC image 24.04 with ROS2 Jazzy.

> **Note**
>
> This is an open Proof of Concept (POC) design and is not officially supported by NXP. The design is enabled with a Vanilla Ubuntu POC layered on top of existing NXP Yocto build. This means ROS2 main installs via `apt install ros2`.
>
> The unsupported software repo can be found here: https://github.com/NXP-Robotics/imx-manifest-navq95

- Cortex-M7 Core: Runs either Zephyr / CogniPilot or NuttX / PX4.
- Communication: High-speed inter-core communication is facilitated using shared memory / OpenAMP RpMsg.

Cortex-M7 realtime software

A range of software platforms can be deployed on the Cortex-M7 core. The following software platforms have been prepared for use with the MR-NAVQ95:

- Zephyr
  - https://www.zephyrproject.org/
  - https://github.com/CogniPilot/zephyr_boards/
- CogniPilot Cerebri (Zephyr based)
  - https://CogniPilot.org/
- NuttX
  - https://NuttX.apache.org/
  - https://NuttX.apache.org/docs/latest/platforms/arm/imx9/boards/mr-navq95b/index.html
- PX4 Autopilot (NuttX based)
  - https://px4.io/

Hardware Specifications

Power and Memory

- **Input Power:** Supports an operating voltage range of 9V to 52V, with a maximum limit of 60V. This easily accommodates 3S to 12S battery configurations, delivering 60W across the 9-20V (3-7S) range, and stepping up to 125W for the 20-52V (8-12S) range.
- **RAM:** Up to 16 GB of LPDDR5 memory.
- **Storage Options:** 64 GB onboard eMMC, an Octal-SPI Flash module, a Micro SD card reader, and PCIe M.2 support for solid-state drives.

Connectivity

- **Standard Ethernet:** One Gigabit RJ45 port supporting Precision Time Protocol (PTP).
- **Automotive Ethernet:** Both 100BASE-T1 and 1000BASE-T1 ports, also featuring PTP.

> **Tip**
>
> 100(0)BASE-T1 is Ethernet over single unshielded twisted pair

- **Wireless Comms:** Powered by the NXP IW612 chip for Wi-Fi, Bluetooth, and Matter support.
- **Cellular Data:** Includes a SIM slot and an M.2 PCIe interface meant for a cellular modem.

Onboard Mobile Robotics Sensors

- TDK ICM-45686 Inertial Measurement Unit (IMU)
- Bosch BMM350 Magnetometer
- Bosch BMP581 Barometer

Physical Interfaces

- USB 2.0 and 3.0 ports.
- Two PCIe M.2 slots (Type M and Type B).
- A 10-pin JTAG SWD debugging header.

Main Board Schematics

The MR-NAVQ95 Main Board serves as the central processing and power distribution hub for the platform. It handles the core i.MX95 compute, wide-input power delivery, and routing to all board-to-board and high-speed interfaces.

For detailed component layouts, pin configurations, and electrical routing, refer to the core system schematics:

- NAVQ95-MAIN

Expansion Capabilities

Building upon this core Main Board, the system is highly adaptable thanks to support for modular add-on boards:

XGMII-based Networking expansion boards:

- NAVQ95-T1SW: T1 Switch utilizing the NXP SJA1110 for six 100BASE-T1 connections and 2x 1000BASE-T1.
- X-MR-NAVQ95E-T1P: T1 Single Phy setup using the NXP TJA1103.

CSI/DSI Vision / Camera expansion board:

- NAVQ95-CAMRP: A 22-pin Raspberry Pi-style connector expansion board for CSI/DSI interfaces.

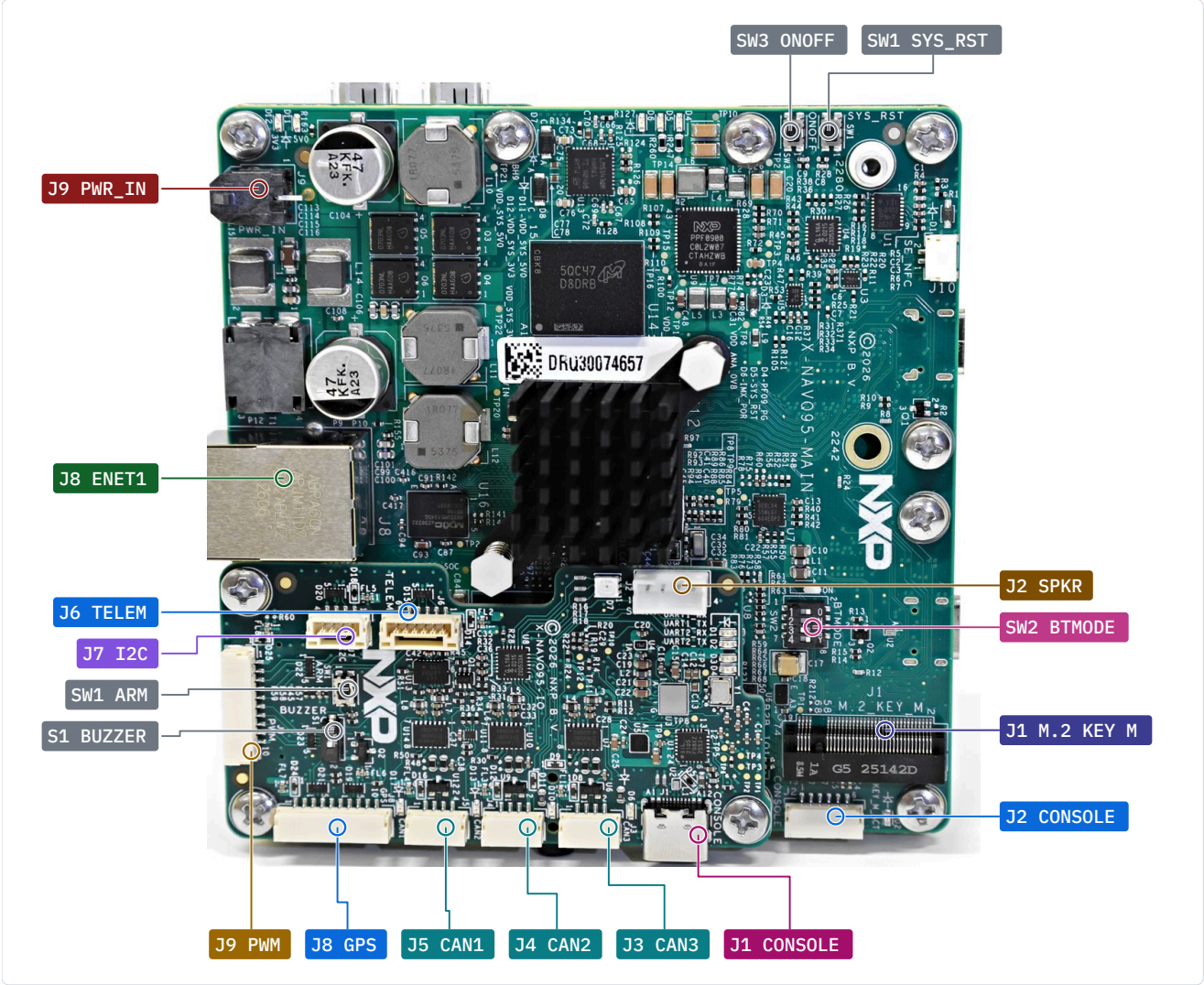
General purpose I/O expansion board:

- NAVQ95-IO: Drone & Rover IO: Uses standard Dronecode connectors for extensive peripheral support:
  - 3x CAN-FD
  - Bosch BMI088 IMU
  - 8x FlexIO/PWM output
  - Dronecode JST-GH 10-pin GPS connector
  - Dronecode JST-GH 6-pin Telemetry connector
  - Dronecode JST-GH 4-pin I2C connector
  - WM8962B Audio codec with:
    - Audio jack
    - 2x PDM Microphones
    - 2x 1W stereo output
  - 2x UART to USB-C for Serial Console

MR-NAVQ95 hardware connector reference

Each card shows three things per pin: the **function**, the **net name in the schematic** and the **i.MX95 pad** the signal ends up on. Parts in between, such as transceivers, level shifters and I/O expanders, are shown too. Pin 1 is the square pad.

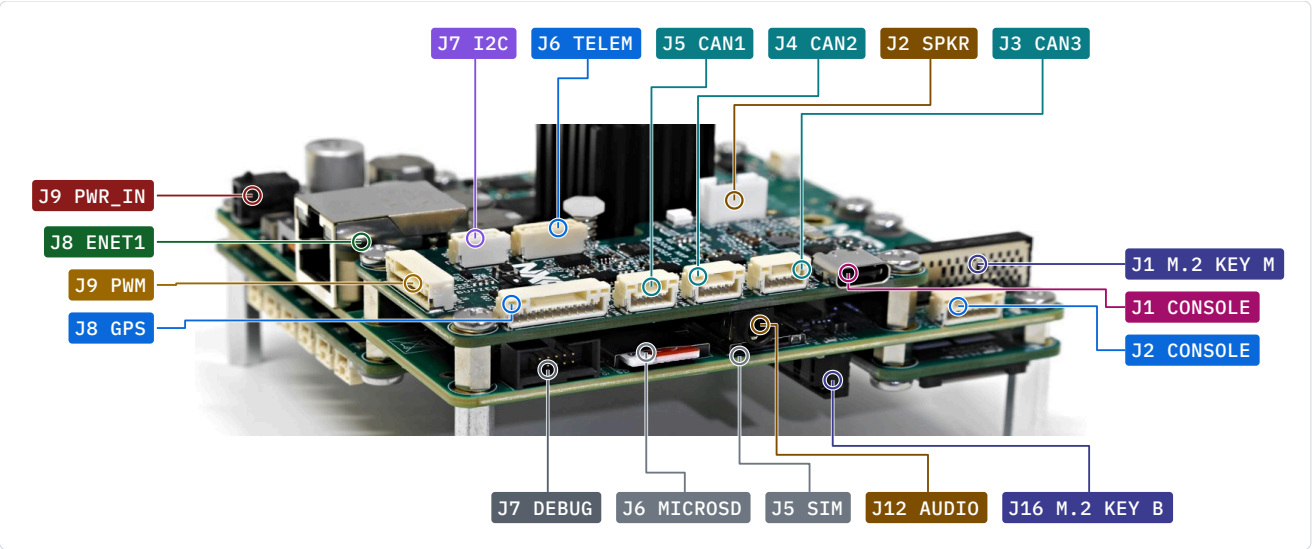
Board overview



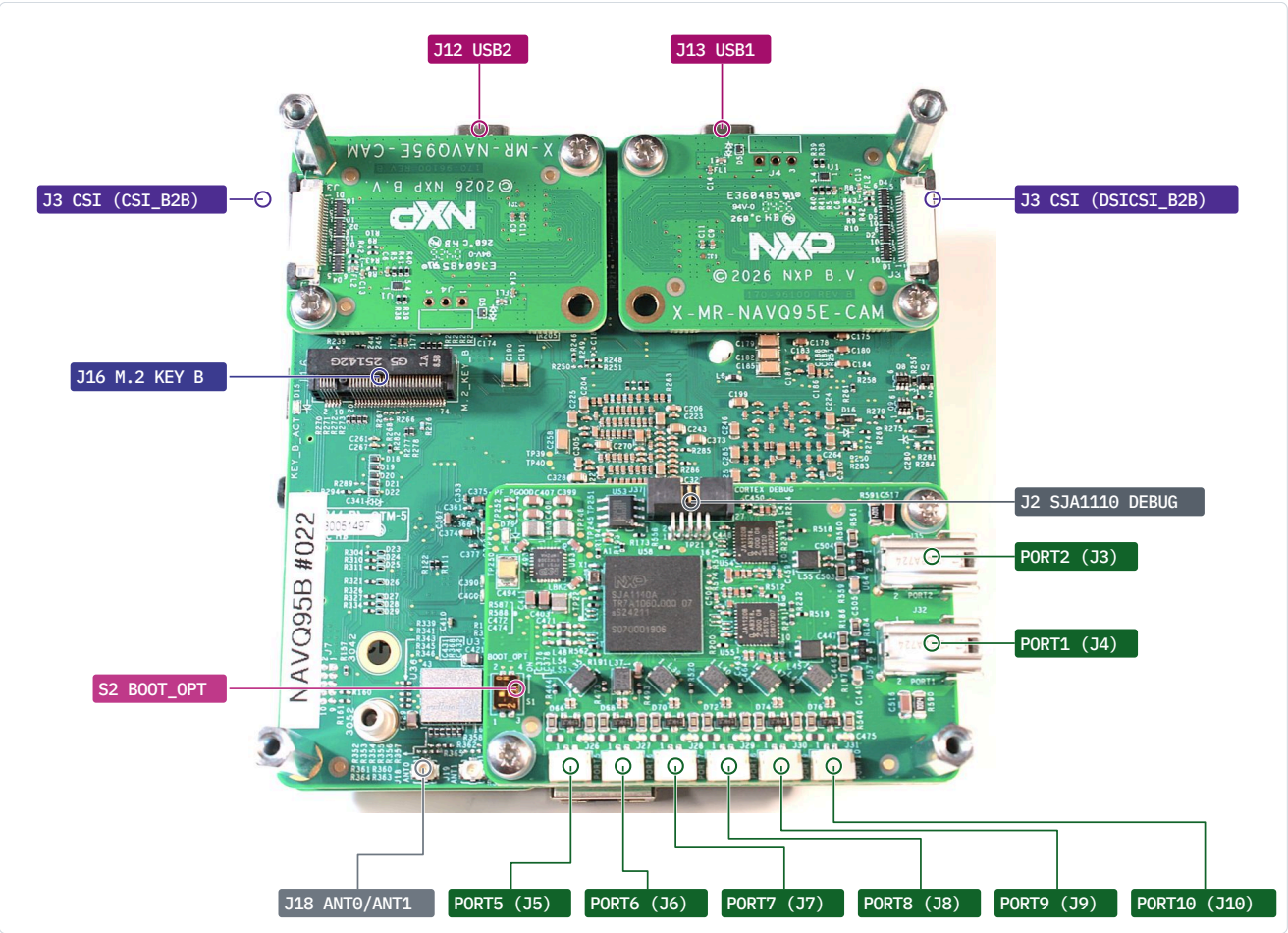
Bottom view: J13 USB1 · J12 USB2. Bottom and front view: J16 M.2 KEY B. Front view: J7 DEBUG · J6 MICROSD. Not visible in the photos: J11 RTC_BAT.

> **Note**
>
> The boot-mode switch (SW2) is explained in the imx-manifest-navq95 README.

Front view. The JST-GH connectors of the IO board are on top. Below them, on the edge of the main board, are the debug header, the microSD slot, the SIM holder and the M.2 Key B slot:



Bottom side. In this photo the T1 Ethernet switch board and two camera boards are fitted. The left camera board is on the CSI_B2B port, the right one on the DSICSI_B2B port; only the DSICSI_B2B port can also drive an RPi DSI display. PORT1 to PORT10 are the switch ports:



Connector index

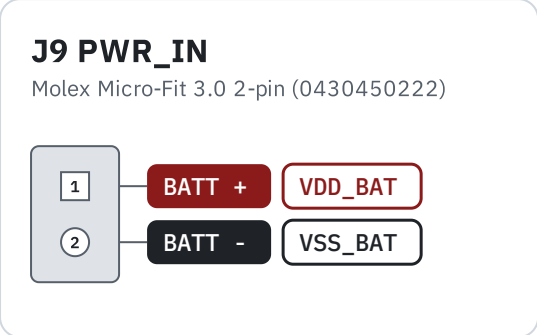
Ref	Function	Connector	Details
Main J1 M.2 KEY M	M.2 Key M slot, PCIe x1 (NPU or NVMe SSD)	M.2 Socket 3 Key M, 2242 / 2280 standoffs, 3.3 V up to 3.5 A	

Ref	Function	Connector	Details
Main J2 CONSOLE	UART1 (Linux console) and UART2 (M7 console) header, DS-009 TELEM layout	JST-GH 1x6 (SM06B-GHS-TB)	
Main J3 B2B	Board-to-board expansion headers	Board-to-board headers	J3/J4 B2B_IO1/IO2 (2x25) for the IO board, J14/J15 B2B_CSI/DSICSI (2x20) for the camera boards, J17 B2B_ETH (2x25) for the T1 boards. Pin lists are on schematic page 25.
Main J5 SIM	SIM card holder for the Key B modem slot	SIM card holder, 6 contacts + card-detect switch, bottom side	
Main J6 MICROSD	microSD card slot (USDHC2)	microSD push-push, bottom side	USDHC2. The board can boot from it when the boot-mode switch selects SD.
Main J7 DEBUG	SWD / JTAG debug header (Arm Cortex 10-pin)	2x5 1.27 mm (Samtec SHF-105-01-L-D-RA)	
Main J8 ENET1	Gigabit Ethernet with PTP	RJ45 with magnetics (Abracon ARJM11D7)	RTL8211 PHY on i.MX95 ENET1, supports IEEE 1588 PTP.
Main J9 PWR_IN	Power input 9-52 V (3S-12S battery)	Molex Micro-Fit 3.0 2-pin (0430450222)	
Main J11 RTC_BAT	RTC backup battery (rechargeable ML2020, RPi 5 compatible)	JST-SH 1.0 mm 2-pin (SM02B-SRSS-TB), top entry	
Main J12 USB2	USB 2.0 Type-C host port	USB Type-C receptacle, bottom side	Host port. 5 V VBUS through an NX20P0477 load switch with over-current protection.
Main J13 USB1	USB 3.0 Type-C, serial download / device port	USB Type-C receptacle, bottom side	Serial-download port of the i.MX95 boot ROM and the port the NavQ95 flasher uses. USB 3.0 and USB 2.0 are both connected.
Main J16 M.2 KEY B	M.2 Key B slot, PCIe x1 with SIM (cellular modem)	M.2 Socket 2 Key B / B+M, 2242 / 3042 / 3052 standoffs, bottom side, 3.3 V up to 5 A	
Main J18 ANT0/ANT1	Wi-Fi 6 / Bluetooth antennas (IW612), ANT0 = J18, ANT1 = J19	2x U.FL	ANT0 = J18, ANT1 = J19. IW612 Wi-Fi 6, Bluetooth and 802.15.4.
IO J1 CONSOLE	USB-C console: CP2105 bridge, UART1 = Linux, UART2 = M7	USB Type-C receptacle (USB 2.0 device)	
IO J2 SPKR	Stereo speaker output (WM8962B class-D, 2x 1 W)	JST-PH 2.0 mm 1x4	
IO J3 CAN3	CAN FD 3 (TJA1463)	JST-GH 1x4, side entry	
IO J4 CAN2	CAN FD 2 (TJA1463)	JST-GH 1x4, side entry	

Ref	Function	Connector	Details
IO J5 CAN1	CAN FD 1 (TJA1463)	JST-GH 1x4, side entry	
IO J6 TELEM	Telemetry UART7 with flow control	JST-GH 1x6, top entry	
IO J7 I2C	I2C6 peripheral connector	JST-GH 1x4, top entry	
IO J8 GPS	GPS: UART5, I2C6, safety switch, LED, buzzer	JST-GH 1x10, side entry	
IO J9 PWM	8x PWM / timer outputs (TPM3 to TPM6), also usable as FlexIO1 pins	JST-GH 1x10, side entry	
IO J12 AUDIO	3.5 mm headset jack, CTIA wiring (WM8962B headphone out + microphone in)	3.5 mm 4-pole TRRS jack with insertion switch, board underside	
CAM J3 CSI	MIPI CSI-2 camera, 4 data lanes, RPi camera compatible	22-pin 0.5 mm FPC/FFC, contacts away from the PCB	
CAM J4 5V	5 V header for a display (not fitted)	1x3 2.54 mm right-angle header, not fitted	
T1S J2 DEBUG	Debug header of the SJA1110 (Arm Cortex-M7 inside the switch)	2x5 1.27 mm	Not the i.MX95 debug port; that is J7 on the main board.
T1S J3 PORT2	1000BASE-T1 port 2	Single-pair automotive connector	TJA1120 PHY on SJA1110 SGMII port 2.
T1S J4 PORT1	1000BASE-T1 port 1	Single-pair automotive connector	TJA1120 PHY on SJA1110 SGMII port 1.
T1S J5 PORT5	100BASE-T1 port 5	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 5.
T1S J6 PORT6	100BASE-T1 port 6	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 6.
T1S J7 PORT7	100BASE-T1 port 7	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 7.
T1S J8 PORT8	100BASE-T1 port 8	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 8.
T1S J9 PORT9	100BASE-T1 port 9	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 9.
T1S J10 PORT10	100BASE-T1 port 10	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 10.

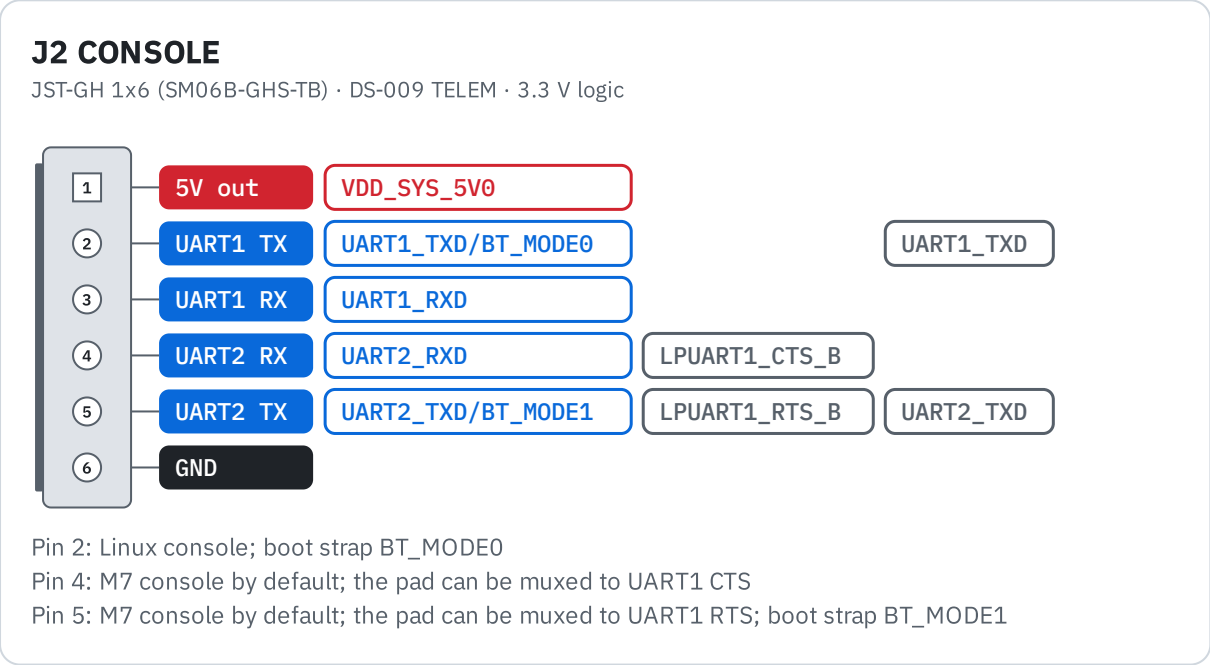
Main board

Main J9 PWR_IN



Pin 1 (BATT +) is the lower pin on the board. Input 9 to 52 V, never more than 60 V. The board can draw 60 W between 9 and 20 V, and 125 W between 20 and 52 V. The input voltage is measured on ADC_IN0 (VIN / 64).

Main J2 CONSOLE



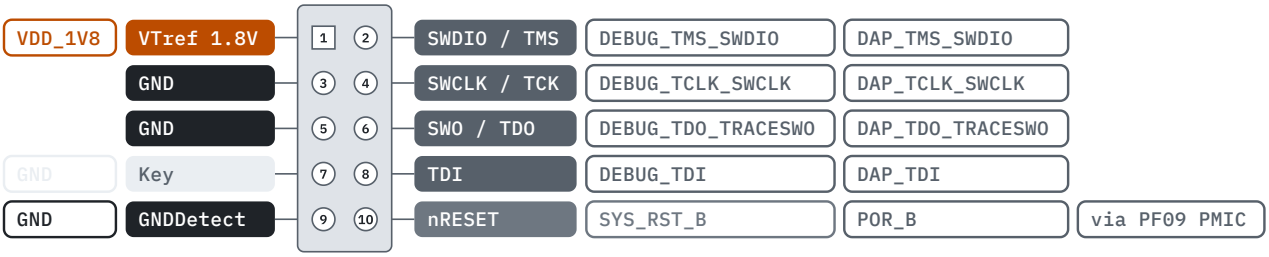
Pins 2 to 5 follow the DS-009 TELEM layout for UART1: 2 = TX, 3 = RX, 4 = CTS, 5 = RTS. By default pin 4 carries UART2 RX and pin 5 UART2 TX (the M7 console). The same pads can be muxed to UART1 CTS and RTS instead, which gives UART1 full DS-009 flow control on this header. UART1 TX and UART2 TX are also the boot-mode straps BT_MODE0 and BT_MODE1. The same two UARTs are also on the USB console port J1 of the IO board, see the figure there.

> **Note**
>
> UART1 TX and UART2 TX are boot straps. Keep them idle while the board resets, or it may boot from the wrong source.

Main J7 DEBUG

J7 DEBUG

2x5 1.27 mm (Samtec SHF-105-01-L-D-RA) · Arm Cortex 10-pin debug · 1.8 V logic (VTref)

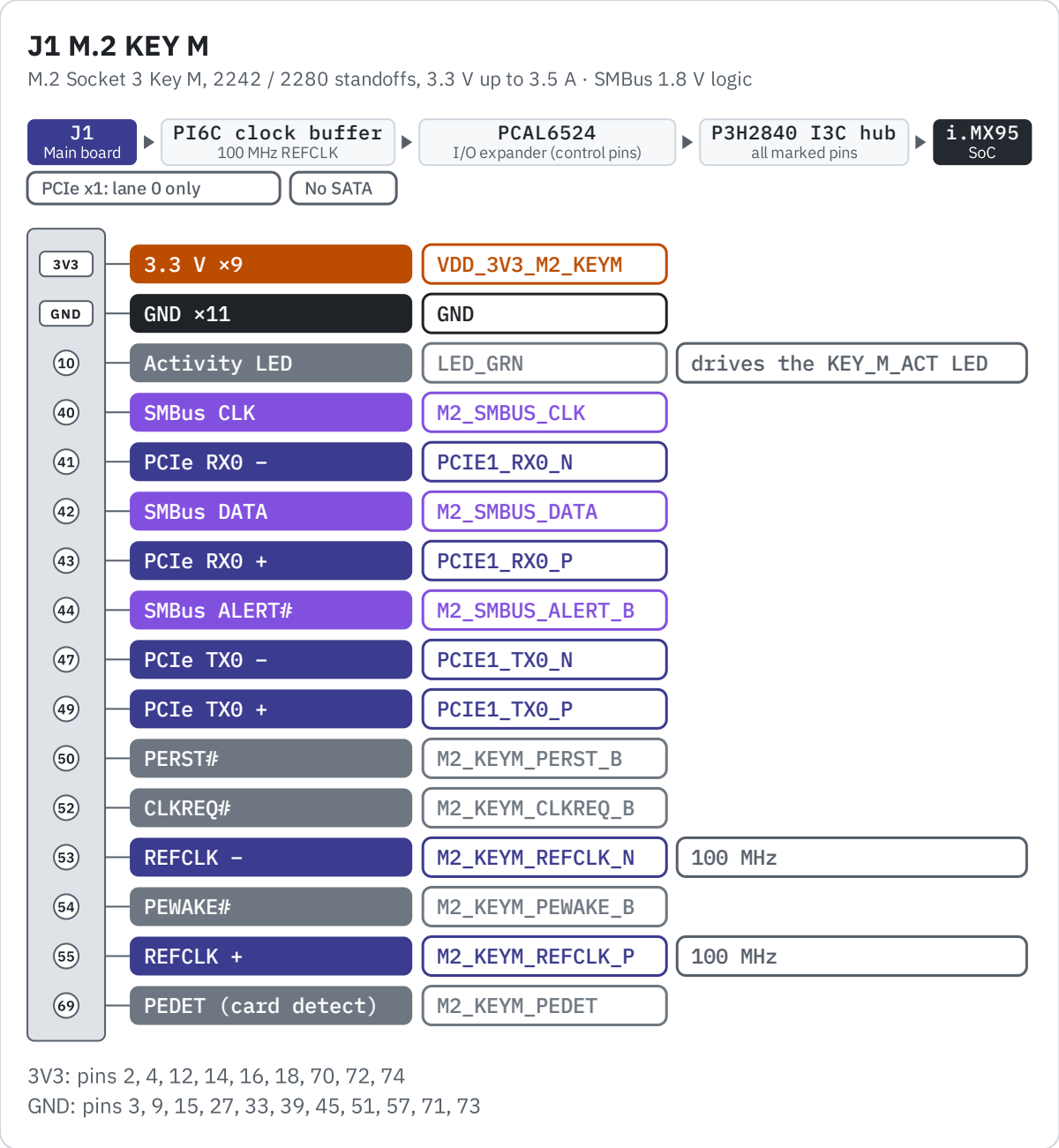


Pin 7: tied to GND
Pin 10: open drain

Standard Arm Cortex 10-pin layout (0.05 inch). Works with J-Link, PyOCD, CMSIS-DAP and similar probes. nRESET goes through a diode to SYS_RST_B, so the probe can only pull it low.

> **Note**
>
> VTref is 1.8 V. Use a debug probe that supports 1.8 V targets.

Main J1 M.2 KEY M



Only the pins below are connected. All other pins of the 75-pin socket are not used: PCIe lanes 1 to 3 (odd pins 5 to 37), SATA, DEVSLP, SUSCLK (pin 68, link resistor not fitted), MFG1/2 and the even pins 6 to 36. The 3.3 V comes through an NX20P5090 load switch that the I/O expander can switch off: 3.5 A continuous, 7 A peaks of 100 µs.

> **Note**
>
> PCIe x1 only, no SATA. An NVMe SSD or NPU module works; a SATA M.2 drive does not.

Main J16 M.2 KEY B

J16 M.2 KEY B

M.2 Socket 2 Key B / B+M, 2242 / 3042 / 3052 standoffs, bottom side, 3.3 V up to 5 A · GPIO, RESET# and SMBus 1.8 V logic

J16 Main board ▸ PI6C clock buffer 100 MHz REFCLK ▸ PCAL6524 I/O expander (control pins) ▸ SIM holder J5 all marked pins ▸ i.MX95 SoC

PCIe x1: lane 0 only

No USB: modem must be a PCIe model

3V3	3.3 V ×5	VDD_3V3_M2_KEYB	
GND	GND ×11	GND	
1	CONFIG_3	M2_KEYB_CONFIG3	
6	FCPO#	M2_KEYB_FCPO_B	full card power off
8	W_DISABLE1#	M2_KEYB_WDISABLE1_B	
10	Activity LED	LED_GRN	drives the KEY_B_ACT LED
20	GPIO_5	M2_KEYB_GPIO5	
21	CONFIG_0	M2_KEYB_CONFIG0	
22	GPIO_6	M2_KEYB_GPIO6	
23	WAKE#	M2_KEYB_WOWWAN_B	wake on WWAN
24	GPIO_7	M2_KEYB_GPIO7	
26	W_DISABLE2#	M2_KEYB_WDISABLE2_B	
30	SIM RST	UIM_RESET	
32	SIM CLK	UIM_CLK	
34	SIM I/O	UIM_DATA	
36	SIM VCC	UIM_PWR	supplied by the module
41	PCIe RX0 −	PCIE2_RX0_N	
43	PCIe RX0 +	PCIE2_RX0_P	
47	PCIe TX0 −	PCIE2_TX0_N	
49	PCIe TX0 +	PCIE2_TX0_P	
50	PERST#	M2_KEYB_PERST_B	
52	CLKREQ#	M2_KEYB_CLKREQ_B	
53	REFCLK −	M2_KEYB_REFCLK_N	100 MHz
54	PEWAKE#	M2_KEYB_PEWAKE_B	
55	REFCLK +	M2_KEYB_REFCLK_P	100 MHz
66	SIM_DETECT		
67	RESET#	M2_KEYB_RESET_B	
69	CONFIG_1	M2_KEYB_CONFIG1	
75	CONFIG_2	M2_KEYB_CONFIG2	

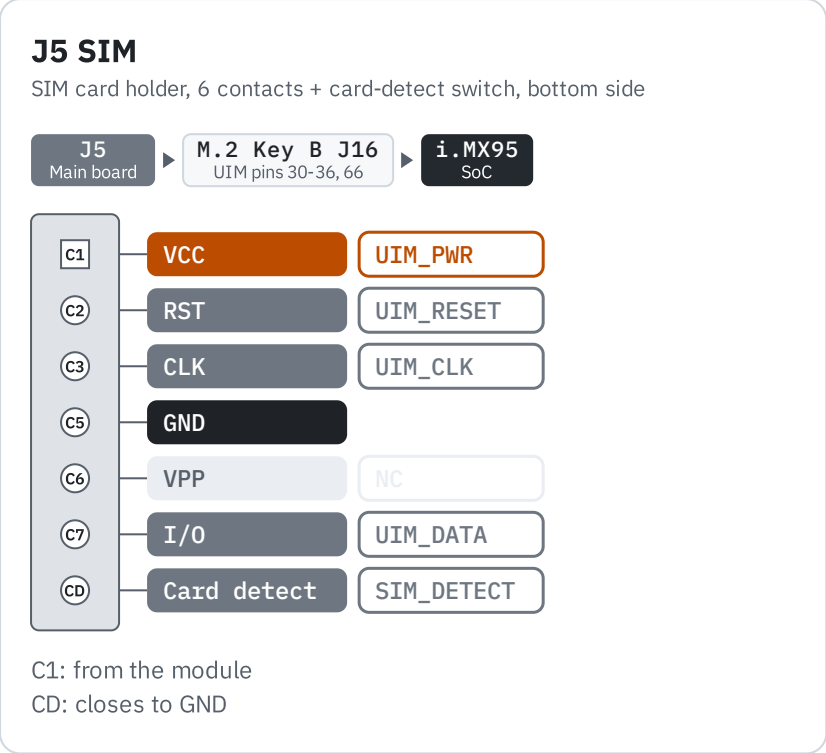
3V3: pins 2, 4, 70, 72, 74
GND: pins 3, 5, 11, 27, 33, 39, 45, 51, 57, 71, 73

Made for a PCIe cellular modem. The USB 2.0 pins (7, 9) and the USB 3.0 / lane 1 pins (29 to 37) are not connected. SMBus (40, 42, 44), SUSCLK (68) and the Bluetooth coexistence lines COEX1/2 (64, 62, to the IW612) are routed, but their link resistors are not fitted. ANTCTL0 to 3, COEX3, DPR, DEVSLP, GPIO_3/4/8 and MFG1/2 are not connected. The UIM pins go to the SIM holder J5. The 3.3 V comes through an NX20P5090 load switch that the I/O expander can switch off: 5 A continuous, 6 A peaks of 100 µs.

> **Important**

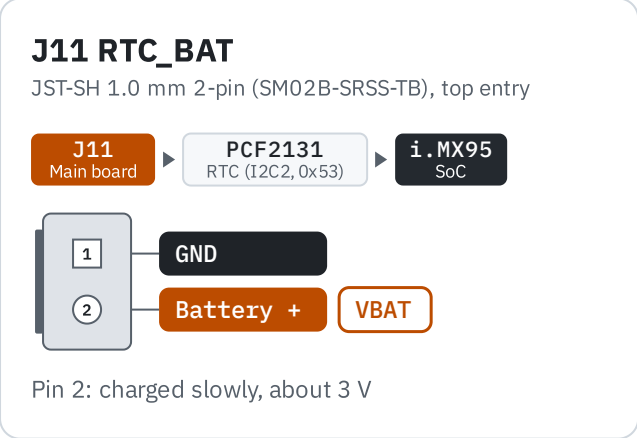
> Buying a 4G or 5G modem? Get one that runs over PCIe. This slot has no USB pins, so a USB modem stays silent. Many modems exist in a USB and a PCIe version with the same name: check the exact SKU or the manual.

Main J5 SIM



Connected directly to the UIM pins of the Key B slot. The modem supplies the SIM voltage on UIM_PWR. The card-detect switch pulls SIM_DETECT low when a card is inserted. All contacts have ESD protection.

Main J11 RTC_BAT



Keeps the PCF2131 real-time clock running while the board has no power. The battery is charged slowly from the 3.3 V rail. This suits a rechargeable ML2020 cell such as the RPi RTC Battery.

> **Warning**
>
> Use a rechargeable ML2020 cell only. A CR2032 or other non-rechargeable cell must not be fitted: the board tries to charge it.

Main other connectors

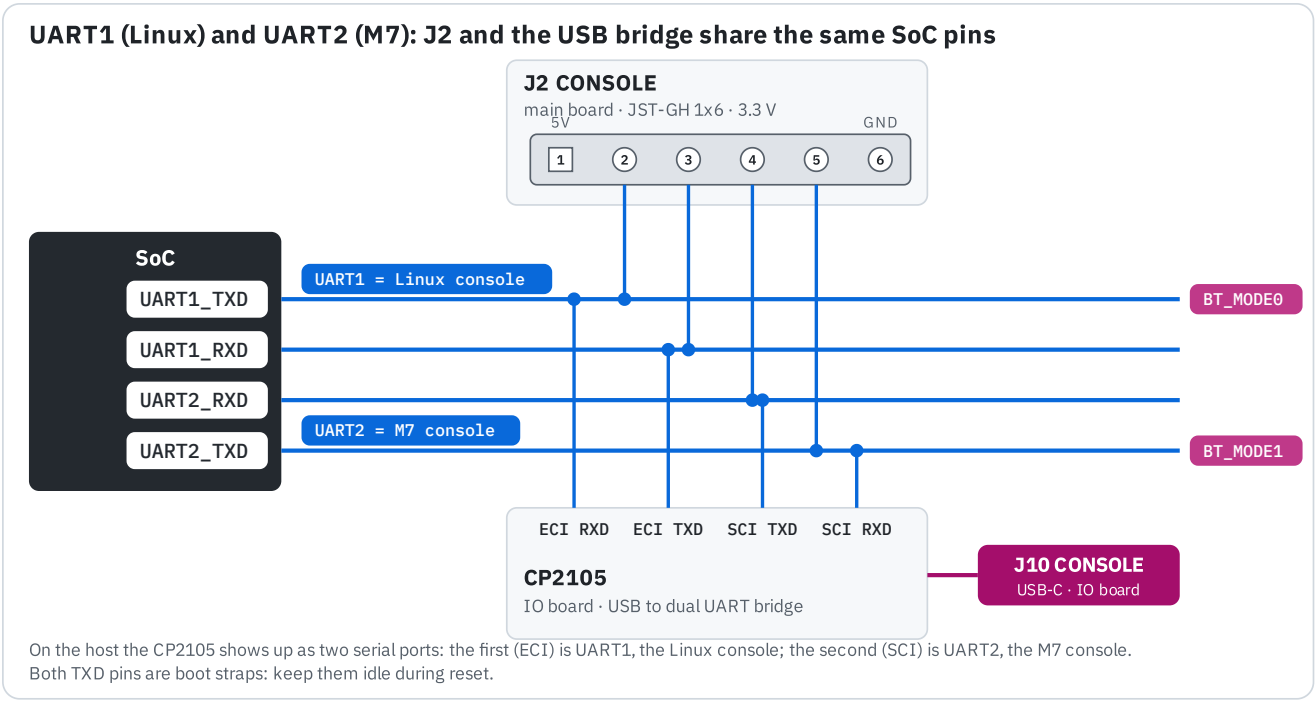
Ref	Function	Connector	Notes
J13 USB1	USB 3.0 Type-C, serial download / device port	USB Type-C receptacle, bottom side	Serial-download port of the i.MX95 boot ROM and the port the NavQ95 flasher uses. USB 3.0 and USB 2.0 are both connected.
J12 USB2	USB 2.0 Type-C host port	USB Type-C receptacle, bottom side	Host port. 5 V VBUS through an NX20P0477 load switch with over-current protection.
J8 ENET1	Gigabit Ethernet with PTP	RJ45 with magnetics (Abracon ARJM11D7)	RTL8211 PHY on i.MX95 ENET1, supports IEEE 1588 PTP.
J6 MICROSD	microSD card slot (USDHC2)	microSD push-push, bottom side	USDHC2. The board can boot from it when the boot-mode switch selects SD.
J18 ANT0/ANT1	Wi-Fi 6 / Bluetooth antennas (IW612), ANT0 = J18, ANT1 = J19	2x U.FL	ANT0 = J18, ANT1 = J19. IW612 Wi-Fi 6, Bluetooth and 802.15.4.
J3 B2B	Board-to-board expansion headers	Board-to-board headers	J3/J4 B2B_IO1/IO2 (2x25) for the IO board, J14/J15 B2B_CSI/DSICSI (2x20) for the camera boards, J17 B2B_ETH (2x25) for the T1 boards. Pin lists are on schematic page 25.

IO expansion board

IO J1 CONSOLE

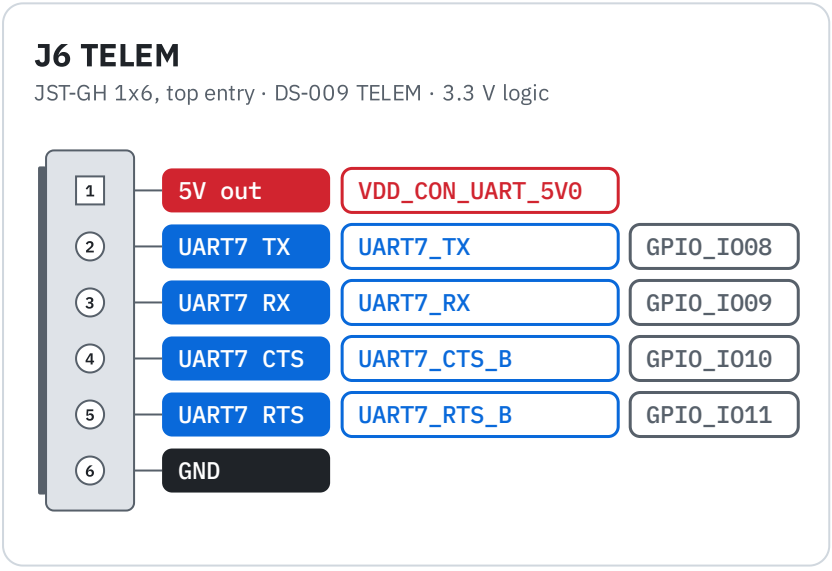
A Silicon Labs CP2105 gives two serial ports on one USB-C connection. On a Linux host they appear as ttyUSB0 (UART1, Linux console) and ttyUSB1 (UART2, M7 console). RX and TX LEDs for both UARTs sit next to the connector. The same two UARTs also go to the J2 header on the main board, see the figure below.

> **Note**
> Port 1 (ECI) is the Linux console, port 2 (SCI) is the M7 console. Do not let the host drive the TX lines while the board resets.



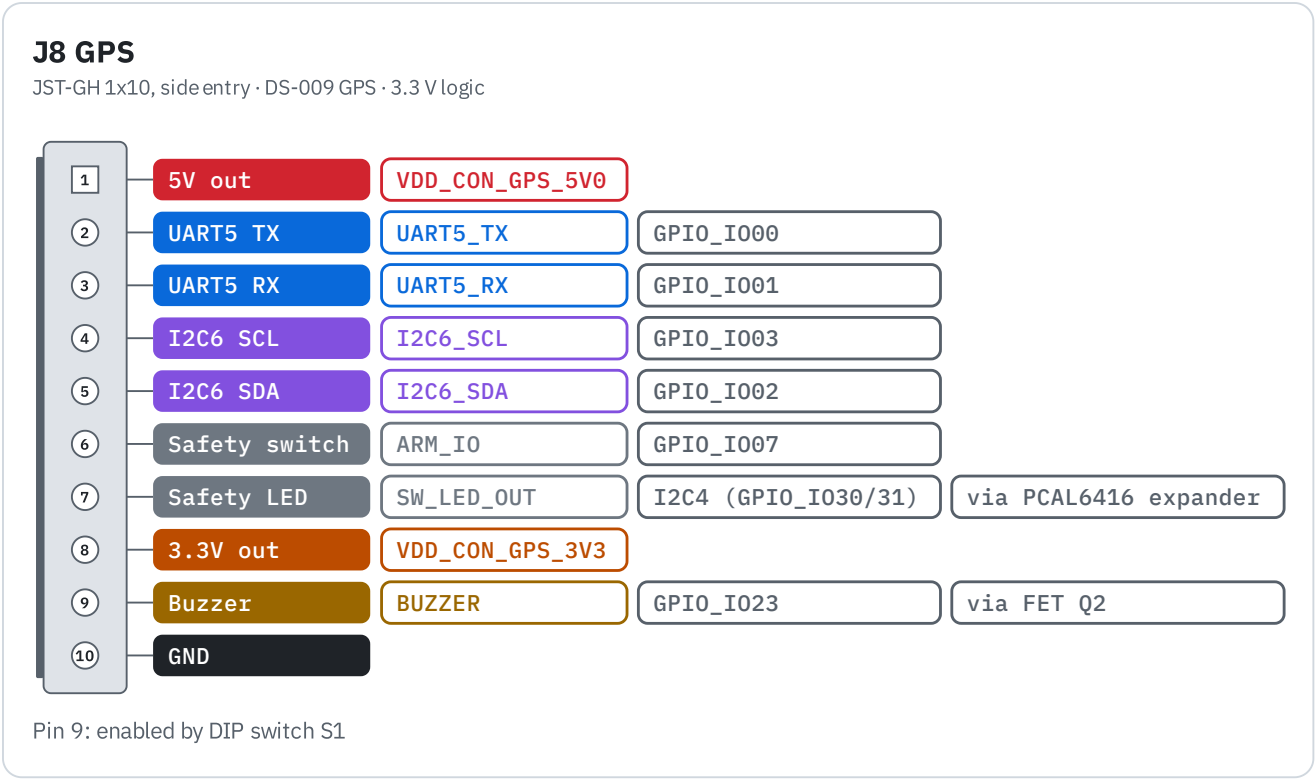
On the host the CP2105 shows up as two serial ports: the first (ECI) is UART1, the Linux console; the second (SCI) is UART2, the M7 console. Both TXD pins are boot straps: keep them idle during reset.

IO J6 TELEM



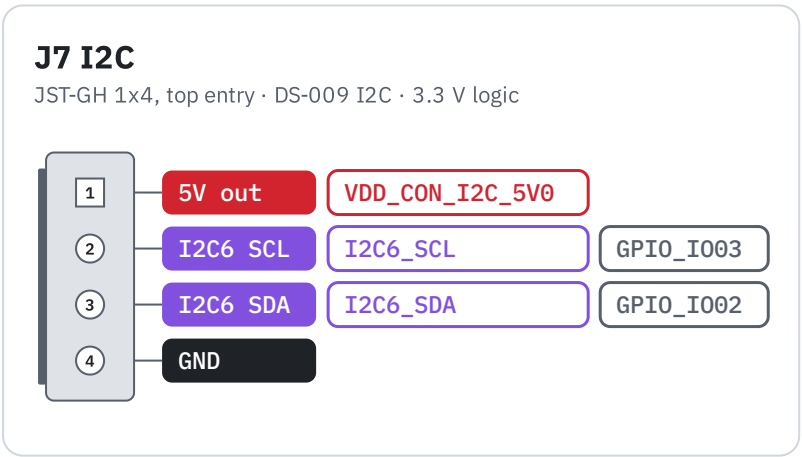
The 5 V output is filtered (VDD_CON_UART_5V0) and has ESD protection.

IO J8 GPS



Standard DS-009 GPS and compass connector. The safety switch input (ARM_IO) is also wired to the ARM button SW1 on the board, and the LED output drives the LED in the switch. The buzzer pin is switched by an N-FET (AO3400A) driven from TPM6_CH1. A buzzer on the board is connected in parallel. DIP switch S1 sits between TPM6_CH1 and the FET: with S1 OFF both buzzers are disconnected and TPM6_CH1 is free for PWM8 on the PWM header.

IO J7 I2C

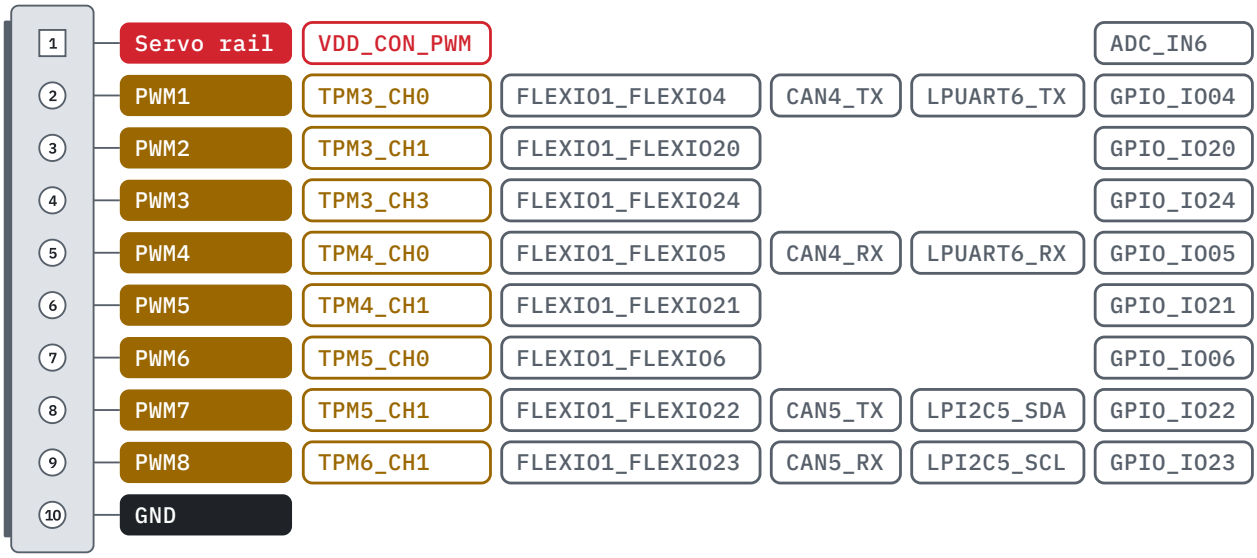


Same I2C6 bus as the GPS connector.

IO J9 PWM

J9 PWM

JST-GH 1x10, side entry · 3.3 V logic outputs



Pin 1: measured on ADC_IN6
Pin 9: shared with the buzzer while S1 is ON

Pin 1 (VDD_CON_PWM) is the servo power rail. It is linked to the board 5 V through R60 (0 Ω, not fitted) and measured on ADC_IN6 (VDD_CON_PWM / 8, up to 12 V). The outputs are 3.3 V logic with ESD protection. PWM8 (pin 9) also drives the buzzer. Every output pad can also be muxed to FlexIO1, for example for DShot or other custom timing protocols. Pins 2 and 5 can also be CAN4 or LPUART6 (TX and RX), pins 8 and 9 CAN5 or LPI2C5 (no pull-ups on the board). CAN4 and CAN5 need an external transceiver.

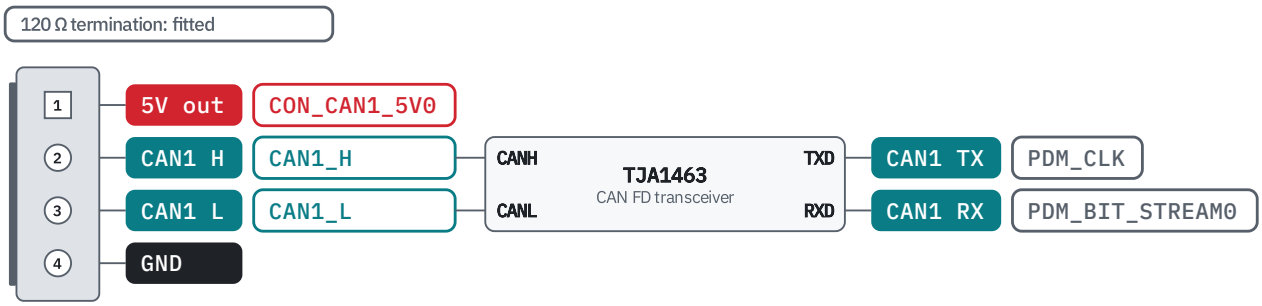
> **Warning**
>
> Do not put more than 5 V on the servo rail (pin 1) while R60 is fitted.

> **Tip**
>
> Need all eight PWM outputs? Switch DIP switch S1 (BUZZER) off. It frees PWM8 from the buzzer.

IO J5 CAN1

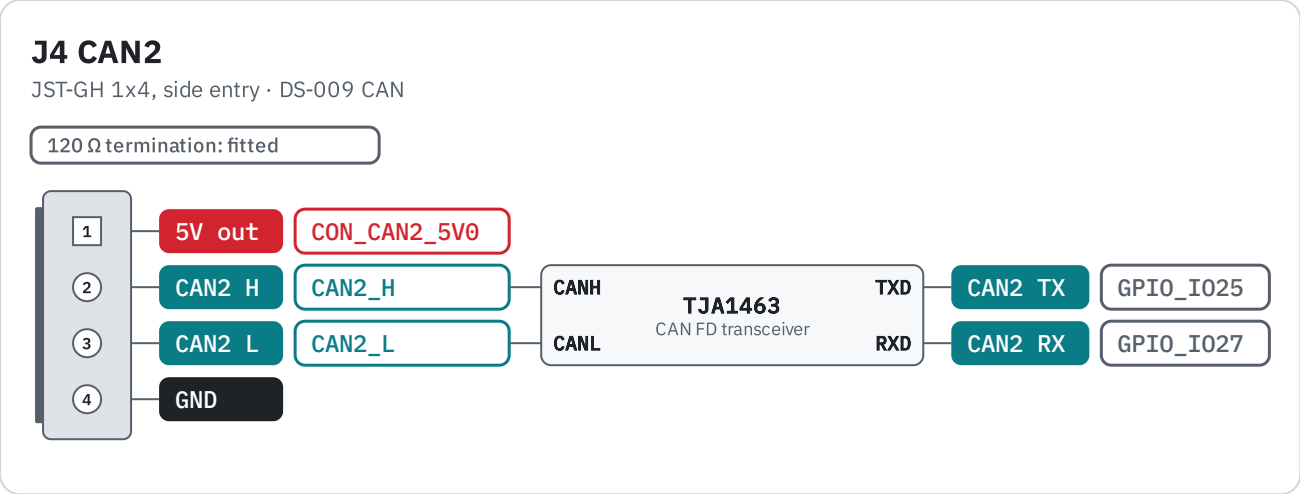
J5 CAN1

JST-GH 1x4, side entry · DS-009 CAN



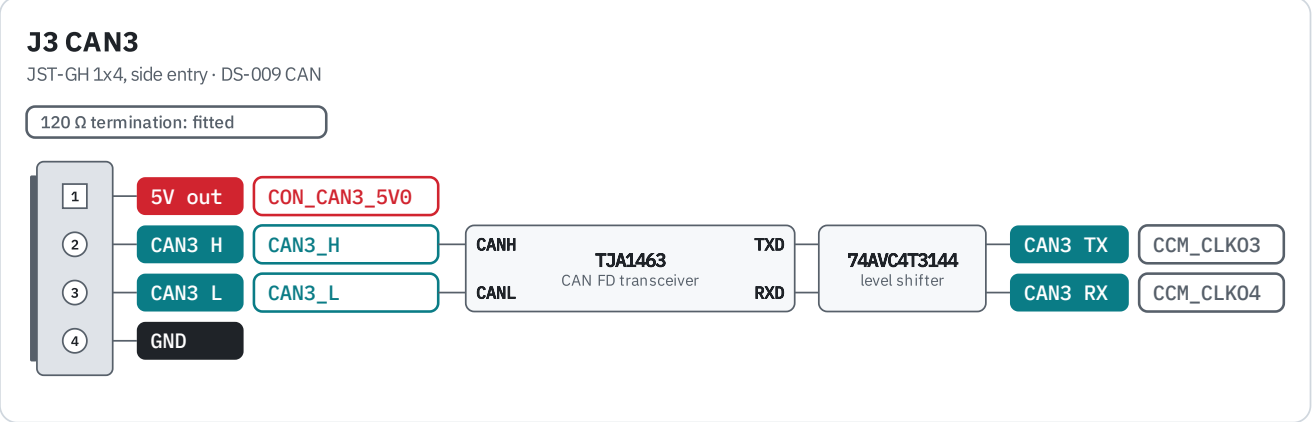
TJA1463 CAN FD transceiver with 5 V supply. The 120 Ω split termination is on the board, so each CAN port is a bus end. CANH and CANL together form the differential bus. TXD and RXD are the logic pins of the transceiver and go to the FlexCAN pads of the i.MX95. The transceiver enable and standby pins are controlled through the PCAL6524 I/O expander.

IO J4 CAN2



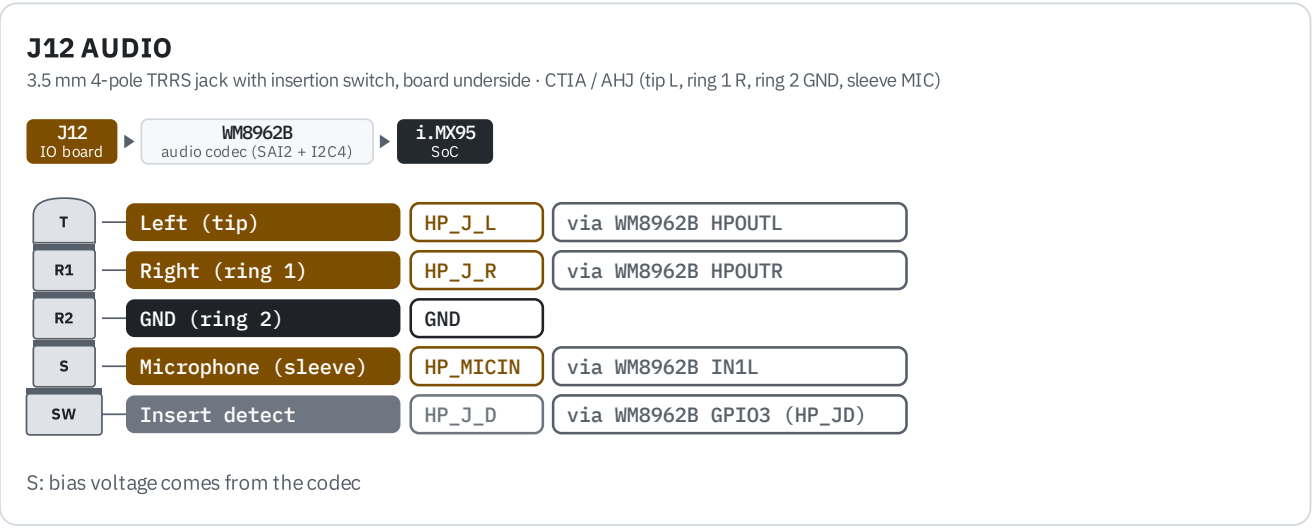
Same circuit as CAN1, with the 120 Ω termination.

IO J3 CAN3



Same circuit as CAN1, with the 120 Ω termination.

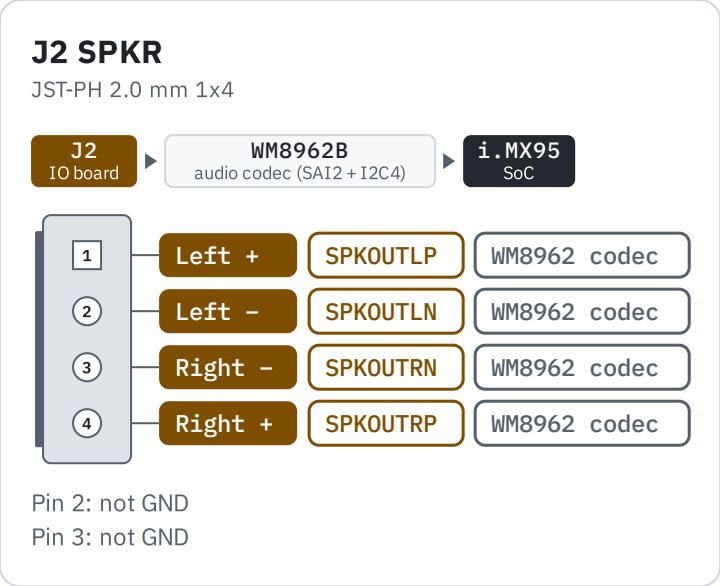
IO J12 AUDIO



Plug contacts listed from tip to sleeve. The switch in the jack drives HP_JD: high with a plug inserted, low without. The codec uses it to detect the plug. The headphone ground reference is the HP_FB sense line of the codec.

> **Note**
>
> CTIA wiring only. OMTP headsets (microphone and ground swapped) do not work.

IO J2 SPKR



Bridge-tied class-D outputs of the WM8962B codec. The 4-pin JST-PH plug of the Adafruit 1669 Stereo Enclosed Speaker Set (2x 3 W, 4 Ω) fits directly. J12 next to it is the headset jack; the board also has two PDM microphones.

> **Caution**
>
> No GND on this connector. Each speaker goes between its – and + pin. Never connect a speaker pin to ground.

Buttons and switches

Ref	Board	Function	Type	Notes
SW1 SYS_RST	Main	System reset button (short press: reset)	Tactile switch, right of the ON/OFF button, hidden under the thermal pad in the photo	Resets the whole board.
SW2 BTMODE	Main	Boot mode DIP switch	4-position DIP switch, top side under the M.2 Key M module	See Power up NavQ95.
SW3 ONOFF	Main	Power button (short press: on, 5 s press: off)	Tactile switch, left of the reset button	Talks to the PMIC ONOFF input.
S1 BUZZER	IO	Buzzer enable DIP switch	1-position DIP switch (CHS-01TA1), next to the ARM button	ON connects TPM6_CH1 to the buzzer FET. OFF disconnects the buzzer and the GPS buzzer pin, so PWM8 on the PWM header is a free channel.
SW1 ARM	IO	Arm / safety switch with LED	Tactile switch	Wired in parallel with the safety-switch and LED pins of the GPS connector.
S1 SMI	T1S	SMI routing switch	1-position DIP switch (CHS-01TA1)	Leave as delivered. Only used when the SJA1110 runs without the host.
S2 BOOT_OPT	T1S	SJA1110 boot option switch	2-position DIP switch (CHS-02TA)	Boot option of the SJA1110; see the SJA1110 documentation before changing it.

Camera expansion board

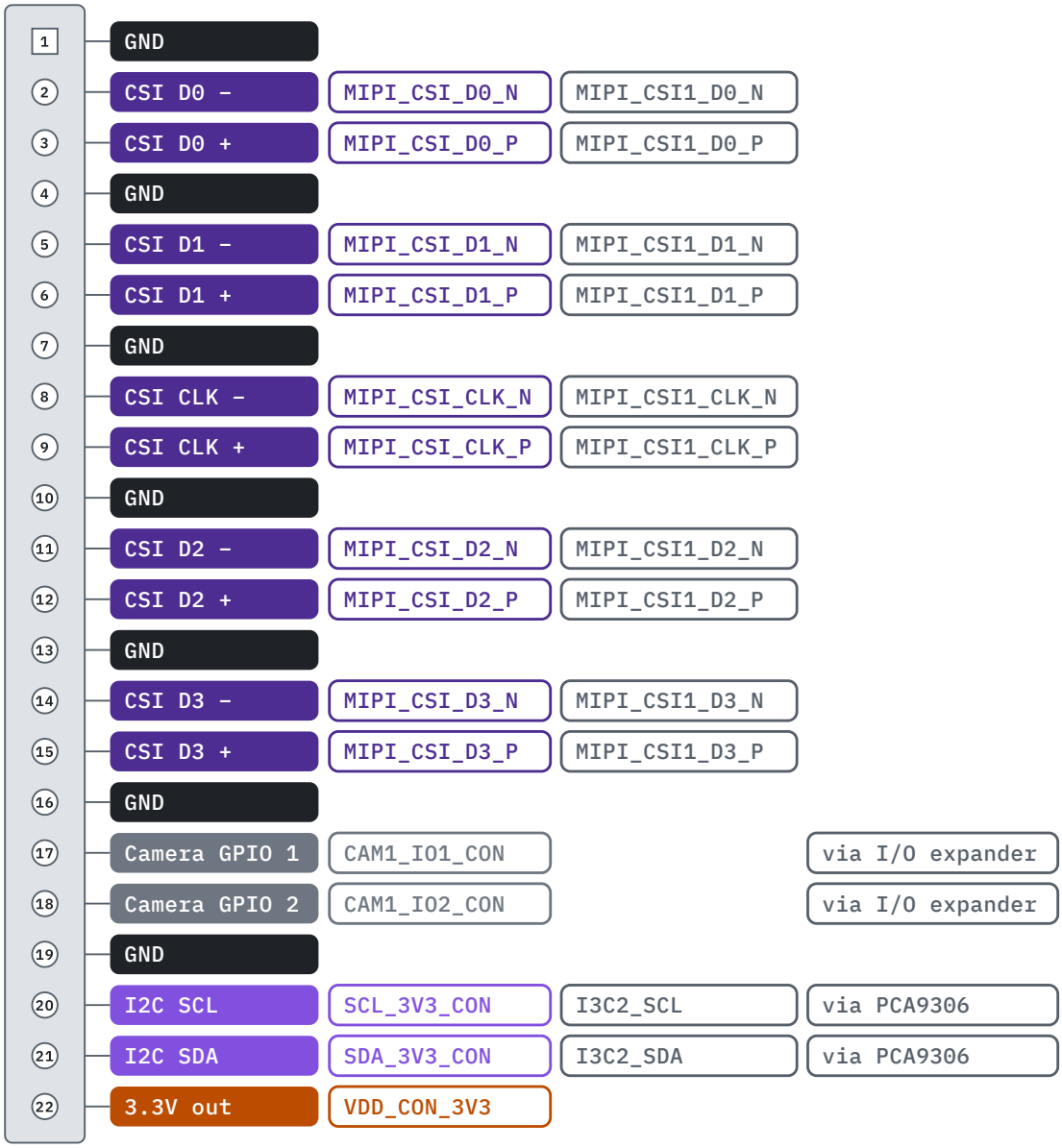
CAM J3 CSI

J3 CSI

22-pin 0.5 mm FPC/FFC, contacts away from the PCB · RPi 22-pin CSI layout · 1.2 V MIPI, 3.3 V I2C and GPIO

J3 CAM board ▸ IP4292 MIPI ESD arrays ▸ PCA9306 I2C level shifter ▸ i.MX95 SoC

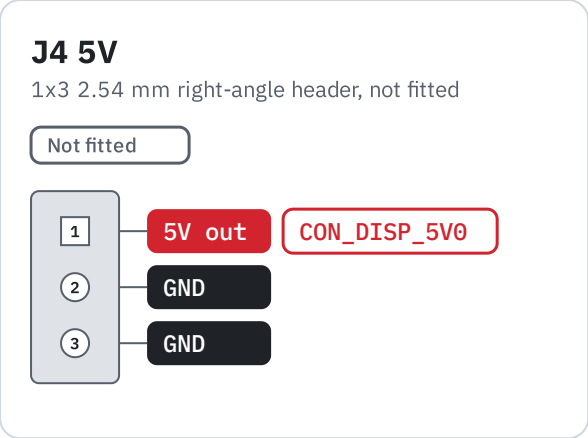
RPi camera modules fit (22-pin FFC) | RPi DSI displays: DSICSI_B2B port only



Pin-compatible with the 22-pin RPi camera connector (22-pin 0.5 mm same-side FFC). Two identical camera boards can be fitted. In the bottom view, the left one sits on CSI_B2B and goes to the i.MX95 MIPI_CSI1 port; the right one sits on DSICSI_B2B and goes to the second port, which is shared between DSI and CSI. The camera I2C is the i.MX95 I3C2 bus, shifted from 1.8 V to 3.3 V by a PCA9306. The two GPIO pins come from the I/O expander on the main board; the exact expander pins and the pad names of the second port were not traced. Whether a camera module works depends on the software image, not on the connector. J4 next to it is a 3-pin 5 V header for a display, not fitted.

> **Tip**
>
> RPi camera modules fit directly with a standard 22-pin FFC. For an RPi DSI display use the camera board on the DSICSI_B2B port; the CSI_B2B port cannot drive a display.

CAM J4 5V



Filtered 5 V from the system rail with ESD protection, meant for a DSI display on the DSICSI_B2B camera board. The silkscreen marks pin 1 as 5V.

T1 Ethernet switch board

The MR-NAVQ95E-T1S board is an NXP SJA1110 automotive Ethernet switch. It is connected to the main board through B2B_ETH on SGMII port 4 of the switch (the host port). Ports 1 and 2 are 1000BASE-T1 through two TJA1120 PHYs on SGMII ports 1 and 2. Ports 5 to 10 are 100BASE-T1 on the switch's own PHYs. SGMII port 3 is not used. The port numbers on the bottom view are the switch port numbers, which also match the silkscreen.

T1S ports and headers

Ref	Function	Connector	Notes
J4 PORT1	1000BASE-T1 port 1	Single-pair automotive connector	TJA1120 PHY on SJA1110 SGMII port 1.
J3 PORT2	1000BASE-T1 port 2	Single-pair automotive connector	TJA1120 PHY on SJA1110 SGMII port 2.
J5 PORT5	100BASE-T1 port 5	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 5.
J6 PORT6	100BASE-T1 port 6	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 6.
J7 PORT7	100BASE-T1 port 7	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 7.
J8 PORT8	100BASE-T1 port 8	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 8.
J9 PORT9	100BASE-T1 port 9	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 9.
J10 PORT10	100BASE-T1 port 10	2-pin single-pair connector	SJA1110 internal 100BASE-T1 PHY, switch port 10.
J2 DEBUG	Debug header of the SJA1110 (Arm Cortex-M7 inside the switch)	2x5 1.27 mm	Not the i.MX95 debug port; that is J7 on the main board.

Notes

- The connectors marked DS-009 on their card (TELEM, GPS, I2C, CAN and the J2 console header) follow the Dronecode DS-009 connector standard, so standard cables fit. The other JST-GH connectors, such as the PWM header, have their own pin order.

- Input 9 to 52 V (never more than 60 V), 3S to 12S batteries. 60 W between 9 and 20 V, 125 W between 20 and 52 V.
- The 5 V pin of every JST-GH connector is a filtered, ESD-protected output of the 5 V system rail (`VDD_CON_*_5V0` nets).
- The servo rail of the PWM connector (`VDD_CON_PWM`) is separated from the 5 V rail by R60, a 0 Ω link that is not fitted. This allows an external servo supply. The rail is measured on ADC_IN6 and must stay below 12 V.
- VIN, 5 V and 3.3 V are measured on ADC_IN0, ADC_IN1 and ADC_IN2.

> **Warning**
>
> Never feed power into a 5 V pin of a JST-GH connector. They are outputs. Power the board through J9 PWR_IN only.

Downloads

This reference as an A4 PDF, and all cards on one double-sided A4 landscape sheet: cheat sheet PDF.

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